

Biology 123 — Master Cheatsheet

Reproduction • Transport • Gas Exchange

Shared Fundamentals Across the Unit

One logic: life runs on exchanges across surfaces. Every rate obeys **Fick's Law**: $\text{Rate} = (A \times \Delta P) / D$. Maximize surface area (A), keep the gradient (ΔP) high, minimize distance (D). Lungs, gills, tracheae, capillaries, stomata, and leaves all follow this rule. Transport systems exist because large bodies cannot rely on diffusion alone — as size grows, **surface-area-to-volume ratio** drops, so cells need help getting supplies and dumping wastes.

- **Diffusion:** Passive high → low concentration movement; no ATP.
- **Osmosis:** Water diffusion across a semipermeable membrane.
- **Gradient:** Difference that drives movement (pressure, concentration, water potential).
- **SA : V ratio:** Small organisms diffuse directly; large ones evolve transport.
- **Mitosis vs. meiosis:** Identical diploids (asexual) vs. recombined haploids (sexual).
- **Gamete:** Haploid sex cell (sperm, egg, pollen, ovule).
- **Zygote:** Diploid cell formed at fertilization.
- **Haploid / Diploid:** 1 vs. 2 sets of chromosomes (n vs. 2n).
- **Cohesion / Adhesion:** Water-water / water-wall — drives xylem.
- **Source / Sink:** Sugar producer (leaf) / sugar user (root, fruit, seed).
- **Endotherm / Ectotherm:** Generates / absorbs heat; endotherms demand more O₂.
- **ATP:** Cellular energy currency made by respiration.
- **Hypotonic / Hypertonic / Isotonic:** Lower / higher / equal solute than cell.
- **Active vs. Passive transport:** With ATP vs. down a gradient for free.

1 • Animal Asexual Reproduction

Core idea: one parent → genetically identical offspring by **mitosis**. Fast, cheap, and reliable in stable environments — but zero genetic variation, so a single pathogen or climate shift can wipe out the whole population. Most often seen in simple animals and in good conditions where rapid population growth is favored.

A. Fission (Binary Fission)

- Parent splits into two **roughly equal** parts; each regenerates missing structures.
- **Examples:** planaria flatworms, sea anemones, protists, many bacteria.

B. Budding

- A **small outgrowth** on the parent grows into a new individual; bud is smaller than parent.
- May detach, or stay attached to form a **colony** (Hydra, corals, sponges).

C. Fragmentation

- A piece of the parent breaks off and regenerates into a whole organism.
- **Examples:** sea stars, sponges, ribbon worms. Small fragments may form **cysts** to survive.

D. Parthenogenesis

- Unfertilized egg develops into a new individual with no sperm contribution.
- **Obligate:** whiptail lizards (all-female species).
- **Facultative:** Komodo dragons, some sharks when isolated.
- **Social insects:** bees — unfertilized eggs → haploid male drones; fertilized → diploid females.

Why asexual works

- No need to find a mate; all offspring can reproduce.
- Best when conditions are **stable** and resources are plentiful.
- Many species switch: asexual in good times, sexual under stress.

	Asexual	Sexual
Parents	One	Two (usually)
Variation	None (clones)	High (meiosis + recombination + fertilization)
Speed / Energy	Fast & cheap	Slower, expensive (mates, gametes, care)
Adaptability	Low — vulnerable to change	High — can evolve with environment
Cell division	Mitosis	Meiosis → fertilization
Best conditions	Stable, resource-rich	Changing, disease-prone

2 • Sexual Reproduction & Plant Reproduction

Sex Determination & Hermaphroditism

- **Gonochoristic:** Sex is **fixed at birth** — most vertebrates, separate males and females.
- **Simultaneous hermaphrodites:** Both male and female organs at once (earthworms, banana slugs); usually **cross-fertilize** to keep variation.
- **Sequential hermaphrodites:** Individuals switch sex during life based on age or social cues.
 - **Protandrous:** Male → female (clownfish: dominant female in anemone).
 - **Protogynous:** Female → male (blue-headed wrasse: largest fish becomes male).
- **Self-fertilization:** Rare — some tapeworms; reduces variation so usually avoided.

Embryonic Sex Determination

- **Genetic (GSD):** Chromosomes decide: XX/XY humans, ZW birds, XO insects.
- **Temperature (TSD):** Incubation temperature sets sex — turtles, crocodilians, some lizards.
- **Environmental:** Density, pH, hormones, or social cues.

r vs. K Selection Strategies

- **r-selected:** many small offspring, little/no care; high mortality (fish, insects, frogs).
- **K-selected:** few large offspring, heavy parental care (mammals, birds, elephants).
- Most species sit on a spectrum between these extremes.

Fertilization & Developmental Modes

- **External fertilization:** Gametes released into water — fish, amphibians, coral. Requires synchrony (**spawning**); huge gamete numbers offset loss.
- **Internal fertilization:** Sperm delivered inside female — higher success, fewer offspring, needs courtship and copulatory organs (penis, claspers) or **spermatophores** (salamanders).

Developmental Modes

- **Oviparous:** Eggs laid externally; embryo develops outside (birds, most reptiles, insects).
- **Ovoviviparous:** Eggs retained inside mother, embryo fed by **yolk only** (some sharks, guppies).
- **Viviparous:** Embryo fed internally by **placenta** (most mammals).

Mammal Groups

- **Monotremes:** Egg-laying mammals — platypus, echidna.
- **Marsupials:** Short gestation + long **pouch** development (kangaroo, opossum).
- **Placental (eutherian):** Longest gestation; highest parental investment.

Reproductive Cycles

- **Estrous cycle:** Unused uterine lining reabsorbed; most mammals.
- **Menstrual cycle:** Uterine lining shed externally; primates.
- **Delayed implantation:** Fertilized egg pauses until conditions are right (bears, some bats, roe deer).
- **Diapause:** Embryonic development pauses during stress.

Plant Reproduction — Flowers, Double Fertilization, Seeds

Whorl	Role	Key parts
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1. Sepals	Protect bud	Green, leaf-like — collectively the calyx
2. Petals	Attract pollinators	Colored, scented — collectively the corolla
3. Stamens (male)	Produce pollen	Anther (makes pollen) + filament (stalk)
4. Carpel/Pistil (female)	Produce ovules	Stigma (sticky), style (tube), ovary (holds ovules)

- **Monoecious vs. Dioecious:** Both sexes on same plant / on separate plants.
- **Pollen grain:** Male gametophyte — 2 sperm + 1 tube cell.
- **Embryo sac:** Female gametophyte inside ovule — 7 cells, 8 nuclei.
- **Pollination:** Transfer of pollen from anther to stigma — wind, water, insects, birds, bats.

Double Fertilization (angiosperm hallmark)

- Sperm 1 + Egg → **Zygote (2n)** → embryo.
- Sperm 2 + 2 polar nuclei → **Endosperm (3n)** — food tissue.
- Ensures food only made when fertilization succeeds.

- **Seed:** Dormant embryo + food + protective coat.
- **Fruit:** Mature ovary — aids seed dispersal (wind, water, animals).
- **Pericarp layers:** Exocarp (skin), mesocarp (flesh), endocarp (pit).

Plant Asexual / Vegetative Reproduction

- **Apomixis:** Clonal seed from mitosis — no fertilization.
- **Runners (stolons):** Strawberry — horizontal stems root at nodes.
- **Rhizomes / tubers:** Ginger / potato — underground stems.
- **Suckers / plantlets:** Raspberry / Mother of Thousands.
- **Grafting & cuttings:** Human-assisted clonal propagation (orchards).

3 • Animal Transport & Human Circulation

Large bodies can't rely on diffusion — a **circulatory system** bulk-moves fluid so every cell is close to an exchange surface. The heart is the pump, vessels are the plumbing, and blood is the delivery fluid carrying O₂, CO₂, nutrients, waste, hormones, and heat.

Circulatory System Types & Vertebrate Hearts

- **Open circulation: Hemolymph** bathes organs directly; low pressure. Most arthropods and mollusks (except cephalopods).
- **Closed circulation:** Blood stays in vessels; heart pumps under pressure. Vertebrates, annelids, octopus/squid.
- Closed systems support higher metabolism, finer control, and more oxygen delivery.

Vertebrate Heart Evolution

- **Fish: 2 chambers** — single circulation; blood loses pressure at gills.
- **Amphibians: 3 chambers** (2 atria, 1 ventricle); some mixing.

- **Most reptiles:** 3 chambers with **partial septum**; can shunt blood (diving turtles).
- **Crocodylians:** 4 chambers but with a mixing vessel.
- **Birds & Mammals: 4 chambers, full separation** of oxygenated and deoxygenated blood — supports endothermy and high activity.

Single vs. Double Circulation

- **Single:** blood through heart once per lap (fish).
- **Double:** pulmonary (to lungs) + systemic (to body) loops — much higher pressure to body.

Human Heart — Cardiac Cycle & Conduction

Cardiac Cycle Phases

- **Systole:** Contraction — blood ejected from chambers.
- **Diastole:** Relaxation — chambers fill with blood.
- **AV valves (tricuspid, mitral):** Between atria & ventricles; close during systole = “lub.”
- **Semilunar valves (aortic, pulmonary):** At ventricle exits; close at end of systole = “dub.”

Electrical Conduction

- **SA node** (pacemaker, right atrium) fires → atria contract.
- **AV node** delays signal ~0.1 s to let atria empty.
- Signal races down **Bundle of His** → **Purkinje fibers** → ventricles contract from bottom up.

- **Stroke volume (SV):** Volume pumped per beat — ~70 mL.
- **Heart rate (HR):** Beats per minute — ~70 at rest.
- **Cardiac output (CO):** HR × SV = **5 L/min** — all your blood every minute.
- **ECG/EKG:** P wave = atrial depolarization, QRS = ventricular depolarization, T = ventricular repolarization.
- **Blood pressure:** Systolic/diastolic (e.g. 120/80 mmHg). Hypertension = ≥ 140/90.
- **Baroreceptors:** Pressure sensors in aorta/carotid → autonomic nerves + hormones adjust HR and vessel tone.
- **Lymphatic system:** Returns leaked plasma (~3 L/day) to veins; failure → **edema**.

Blood Vessels, Blood Cells & Gas Carriage

Vessels

- **Arteries:** Thick muscular walls; carry blood **away** from heart under high pressure.
- **Arterioles:** Small arteries that regulate flow into capillaries (vasoconstriction/dilation).
- **Capillaries: One cell thick** — the actual exchange site.
- **Venules / Veins:** Thinner walls, lower pressure, **valves** prevent backflow; return blood to heart.

Blood Composition

- **Plasma (~55%):** Water, proteins, ions, nutrients, wastes, hormones.
- **Red blood cells (erythrocytes):** Carry O₂ via **hemoglobin**; no nucleus in mammals.
- **White blood cells (leukocytes):** Immune defense.
- **Platelets:** Cell fragments that clot bleeding.

- **O₂ transport:** ~98% bound to **hemoglobin**; ~2% dissolved in plasma. Each Hb carries up to 4 O₂.
- **CO₂ transport:** ~70% as **bicarbonate (HCO₃⁻)** in plasma; ~23% on Hb; ~7% dissolved.
- **Bohr effect:** Low pH / high CO₂ / high temp → Hb **releases** O₂ where tissues need it most.
- **Myoglobin:** Single-unit oxygen reserve in muscle; critical in diving mammals (seals, whales).
- **Respiratory pigments: Hemoglobin** (Fe, red) — vertebrates. **Hemocyanin** (Cu, blue) — mollusks, arthropods. **Hemerythrin** (Fe, violet) — some worms.
- **Clotting cascade:** Platelets + fibrinogen → fibrin mesh → scab.

4 • Plant Transport & Stress Responses

Xylem moves water and dissolved minerals **up** from soil to leaves; **phloem** moves sugars from **source** to **sink**. Most plant transport is passive — powered by evaporation from leaves (transpiration) and osmotic gradients generated by active sugar loading. Plants never developed a pump, so they rely on physics of water.

Vascular Tissues — Xylem vs. Phloem

	Xylem	Phloem
Transports	Water + dissolved minerals	Sugars (sucrose), amino acids, hormones
Direction	One-way: up (roots → leaves)	Bidirectional: source → sink
Cell state	Dead at maturity (hollow tubes)	Living cells

Conducting cells	Tracheids (all plants) + Vessel elements (angiosperms)	Sieve-tube elements — no nucleus, kept alive by companions
Support cells	—	Companion cells — ATP-powered loading/unloading
Driver	Transpiration pull (passive)	Pressure-flow (active loading needed)
Walls	Lignified — also provide structural support	Thin walls; sieve plates with pores

Mechanisms of Movement

Transpiration – Cohesion – Tension (xylem)

- Evaporation of water at the **stomata** creates negative pressure (tension).
- Cohesion**: water-to-water hydrogen bonds keep the column intact.
- Adhesion**: water sticks to xylem walls — helps hold column up.
- The entire water column is **pulled** up like drinking through a straw — from root hair to leaf.
- Explains why tall trees (redwoods, 100+ m) can lift water against gravity.

Water Potential (ψ)

- Water moves from high ψ → low ψ.
 - **Turgid**: Swollen with water — healthy plant posture.
 - **Flaccid**: Cell loses turgor — wilting.
 - **Plasmolysis**: Cell membrane pulls away from wall in hypertonic solution.

Plant Stress Responses

Drought / Heat

- **Cavitation**: Air bubble (embolism) breaks xylem column — disaster for water movement.
- **Stomatal pits & trichomes**: Sunken stomata + leaf hairs trap humid air and block drying wind.
- **Thick cuticle**: Waxy waterproof leaf coating.
- **CAM photosynthesis**: Stomata open only at **night** to save water — cacti, succulents, pineapple.
- **C₄ photosynthesis**: Separates CO₂ capture and Calvin cycle spatially; efficient in heat.
- **Leaf drop (deciduous)**: Shed leaves in drought or cold to cut water loss.

Pressure-Flow Hypothesis (phloem)

- At **source** (mature leaf), sugar is **actively loaded** into sieve tubes.
- Water follows by osmosis → high pressure at source.
- At **sink** (root, fruit, growing tip), sugar is unloaded; water leaves → low pressure.
- Sap flows high → low pressure, carrying sugars along.
- Source/sink roles can **switch seasonally** — a root can store sugar in fall and release it in spring.

Root Uptake

- **Root hairs**: Increase surface area for absorption.
- **Endodermis**: Innermost root cortex layer — the “bouncer.”
- **Aeroparian strip**: Waxy **suberin** band that forces all water through cell membranes for **selective** mineral uptake.
- **Mycorrhizae**: Fungal root partners — boost water/phosphorus uptake.

Flooding / Low Oxygen

- **Lenticels**: Spongy pores in bark allowing gas exchange even when bark is wet.
- **Aerenchyma**: Air-filled spongy tissue piping O₂ from leaves down to submerged roots (rice, water lilies).
- **Pneumatophores**: “Knee” roots sticking above water (mangroves, bald cypress).
- **Adventitious roots**: Roots that sprout from non-root tissue (stems) under stress.

Cold & Salt

- **Antifreeze solutes**: Raise solute concentration inside cells to prevent ice damage.
- **Salt glands**: Mangroves and other halophytes secrete excess salt from leaves.

5 • Gas Exchange Fundamentals & Animal Systems

Rate = (A × ΔP) / D. Every exchange structure — gills, lungs, tracheae, alveoli, skin — maximizes **surface area** (folded, branched, lamellar), maintains the **gradient** by ventilating and circulating, and keeps **distance** tiny by being one cell thick and always **moist** (gases must dissolve to diffuse).

Key Vocabulary & Ventilation Methods

- **Gas exchange**: Passive diffusion of O₂ in and CO₂ out across a moist surface.
- **Respiration — 3 meanings**: **External**: breathing. **Internal**: capillary-to-cell exchange. **Cellular**: ATP production from O₂.
- **Aqueous requirement**: Exchange surfaces must stay moist; that is why lungs are internal and amphibian skin must stay wet.
- **Partial pressure**: Pressure of one gas in a mixture — what drives diffusion.
- **Counter- vs. co-current**: Flow direction of blood vs. water/air — determines efficiency.

Ventilation Methods

- **Buccal pumping**: Mouth/throat gulp pushes air/water past exchange surface (fish, amphibians).
- **Ram ventilation**: Swim fast with mouth open — water forced across gills (tuna, sharks).
- **Costal ventilation**: Ribs expand chest cavity — negative pressure pulls air in (reptiles, mammals).
- **Tidal ventilation**: Air moves **in and out the same path** — mammals; not 100% efficient (residual volume).
- **Countercurrent (fish gills)**: Blood flows **opposite** to water across lamellae → gradient stays high the whole way → ~80–90% O₂ extraction.
- **Cross-current (birds)**: One-way airflow through parabronchi; even better at low O₂ (high altitude).

Exchange Structures Across the Animal Kingdom

Group	Structure	Key features
Protists / Small invertebrates	Cell surface	Direct diffusion — works only at small sizes.
Flatworms, sponges	Body wall	Flat/thin body keeps every cell near the surface.
Fish	Gills	Lamellae (massive surface area), operculum cover, countercurrent flow.
Amphibians	Lungs + skin	Simple sac lungs + thin moist vascular skin = cutaneous respiration .
Insects	Tracheal system	Spiracles → tracheae → tracheoles deliver air directly to cells — bypasses blood.
Reptiles	Lungs (sac-like)	More folded than amphibian lungs; rib-driven ventilation.
Birds	Lungs + air sacs	Parabronchi — one-way, cross-current airflow; extremely efficient at altitude.
Mammals	Lungs	Bronchi → bronchioles → alveoli (hundreds of millions); diaphragm-driven tidal ventilation.

Human Breathing Control

What controls breathing?

- Rate is set mainly by **CO₂** / **pH**, NOT O₂.
- Holding your breath fails because rising CO₂ → falling blood pH triggers the urge to breathe.
- **Chemoreceptors** in medulla, aorta, carotid sense blood CO₂/pH.
- **Medulla oblongata** signals diaphragm + intercostals.

Breathing Mechanics

- **Inhale:** diaphragm contracts (flattens), ribs lift → chest volume ↑ → pressure ↓ → air rushes in.
- **Exhale:** diaphragm relaxes, chest volume ↓ → pressure ↑ → air pushed out (passive at rest).
- **Bohr effect** (p. 3) makes O₂ unload automatically at hard-working muscles.
- Residual volume: lungs never fully empty — reason tidal ventilation is less efficient than one-way.

6 • Plant Gas Exchange, Leaves & Unit Summary

Plants need CO₂ in for **photosynthesis** (daytime), plus O₂ in for **cellular respiration** (always, but net uptake at night), and O₂ out during the day. All exchange is **passive diffusion** through specialized pores — **stomata** in leaves and young stems, **lenticels** in woody bark.

Stomata, Guard Cells & Leaf Tissues

- **Stomata:** Adjustable pores, usually on leaf underside to reduce water loss; sole path for CO₂ in.
- **Guard cells:** Kidney-shaped pair flanking each stoma; open/close by **turgor** driven by K⁺ and water.
 - Moist conditions: guard cells turgid → **open**.
 - Heat / drought: guard cells flaccid → **closed** (saves water, limits CO₂).
- **Cuticle:** Waxy waterproof layer over epidermis — forces gas exchange through stomata.
- **Lenticels:** Raised pores in bark of woody plants for gas exchange.
- **Epidermis:** Protective single cell layer on leaf top and bottom; transparent.

Mesophyll (middle of leaf)

- **Palisade mesophyll:** Tightly packed column cells on the upper side — main **photosynthesis** site.
- **Spongy mesophyll:** Loose cells with big air spaces on the lower side — gases diffuse to every cell.

CAM Photosynthesis (water savers)

- Stomata open at **night** — CO₂ stored as **malic acid** in vacuoles.
- Closed by day — saves water in heat; stored CO₂ released internally to fuel photosynthesis.
- Examples: cacti, pineapples, succulents, agave.

C₄ Photosynthesis

- Spatial split: CO₂ fixed in mesophyll, Calvin cycle in bundle sheath.
- Efficient in hot, bright conditions — corn, sugarcane.

Leaf Morphology & Modifications

Part / Type	Description / Example
Blade (lamina)	Wide flat photosynthetic surface — catches light.
Petiole	Stalk joining blade to stem — positions leaf.
Stipules	Small leaf-like outgrowths at base of petiole.
Simple leaf	One undivided blade (maple, oak).
Compound leaf	Blade split into leaflets (palmate: horse chestnut; pinnate: ash, rose).
Bracts	Petal-like modified leaves; often mistaken for petals (poinsettia red “flowers”, dogwood).
Spines	Defense modification (cactus spines are modified leaves).
Tendrils	Thin climbing structures (peas, grapes).
Storage leaves	Fleshy water/nutrient stores (onion scales, succulents).
Insectivorous leaves	Trap & digest insects in nutrient-poor soil — Venus flytrap, sundew, pitcher plant.

Unit Summary — Patterns to Remember

- **Exchange obeys Fick’s Law:** Every system maximizes A, maintains ΔP, minimizes D — gills, lungs, alveoli, capillaries, stomata, leaves.
- **Water is universal:** Gas exchange needs moist surfaces; plant transport runs on water columns; fertilization often happens in water.
- **Cost drives strategy:** Asexual is cheap but inflexible; sexual is expensive but adaptive. Open circulation is cheap; closed is powerful. Diffusion is free; active transport costs ATP.
- **Variation wins long-term:** Meiosis (animals), double fertilization (plants), seed dispersal — all deliver the variation populations need to survive change.
- **Structure fits function:** Alveoli are tiny/many; lamellae are stacked; capillaries are one cell thick; leaves are flat; xylem cells are dead hollow tubes — geometry always serves the job.